

Ziliang (Johnson) Zhang

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Education

University of California, Riverside

Doctor of Philosophy, Electrical and Computer Engineering

09/2022 ~ (Exp.) 12/2026

Research focus: Real-time AI systems, specialized in recommender systems and AI wearables

University of California, Riverside

09/2019 ~ 03/2021

Master of Science, Computer Engineering

Study area: real-time systems, computer architecture, on-device machine learning

University of California, San Diego

Bachelor of Science, Cognitive Science: Human-Computer Interaction

08/2015 ~ 03/2019

Study area: AI, parallel algorithms, data analytics, full-stack software development

Work Experience

AI/ML Engineer Intern

LinkedIn

06/2026 ~ 09/2026

- Researched industry-scale Generation Recommendation (GR) which serves 1.7B monthly traffic.
- Tackled LinkedIn GR model cold start problem and achieved significant increase of offline metrics.

GPU Coder Internship

MathWorks

06/2025 ~ 09/2025

- Developed GPU optimization in ML compiler/IR to speed up the CUDA production code by 17x.
- Researched AI algorithm on CUDA codegen from MATLAB and increased 76% GPU Utilization.

Full-Stack Software Developer

Verizon

07/2021 ~ 09/2022

- Developed micro-service APIs with Spring Framework, Redis, and Angular for 70M monthly traffic.

Graduate Researcher

ESCALab, UCR

09/2020 ~ 03/2021

- Prototyped XR with TPU for pose estimation at 30 FPS stream and 83% less end-to-end latency.
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Highlight Publications

Z. Zhang, H. Kim, C. Liu. RELAXR: Real-time Execution Framework for Latency-aware AI-native Extended Reality. In Submission

- Prototyped agentic AI glasses and improved real-time latencies in both XR and AI pipelines by 56%.

Z. Zhang, C. Liu, H. Kim. MATER: Mutually Aware Framework for Teleoperated-robot with Extended Reality. ACM SIGMETRICS 2026 (SIGMETRICS'26)

- Overhauled XR teleoperation system with ML approximation that shortens completion time by 48%.

Y. Chen*, **Z. Zhang*** (Co-first), H. Kim, C. Liu. REFiNE: Reward and Energy Fairness for Robust Multi-Agent Coordination in RL-driven Robots. In Submission

- Achieved fairness in Multi-agent Reinforcement Learning scenarios that prolongs runtime by 22%.

A. Bukhari, B Mamo, M. S. Hossain, **Z. Zhang**, M. Karimi, D. Enright, P. Manosalva, H. Kim. Low-Cost Sensing and Classification for Early Stress and Disease Detection in Avocado Plants (IEEE IoT-J'25)

- Proposed disease detection methods using LSTM and DNN algorithm to achieve 86% accuracy.

Z. Zhang, Z. Li, H. Kim, C. Liu. BOXR: Body and head motion Optimization framework for eXtended Reality. IEEE Real-Time Systems Symposium (RTSS'24)

- Discovered the Camera-to-Display latency and optimized XR framework to reduce 63% latency.

Z. Zhang, C. Liu, H. Kim. MII: A Multifaceted Framework for Intermittence-aware Inference and Scheduling. ACM International Conference on Embedded Software (EMSOFT'24)

- Delivered layer-wise checkpointing and real-time scheduling to increase 39% successful jobs.

Z. Zhang, C. Liu, H. Kim. Toward a Predictive eXtended Reality Teleoperation System with Duo-Virtual Spaces. Southern California Robotics Symposium (SCR'24)

- Proposed XR-robot teleoperation predictive display that reduces teleoperation error by 73%.

Y. Zhuy, H. Alrashid, S. Bai, C. Zhang, **Z. Zhang**, Z. Qu, R. Y. Ali, A. Magdy. On the Ecosystem of High-Definition (HD) Maps. IEEE 40th ICDEW (ICDEW'24)

- Analyzed over 53 state-of-the-art approaches in HD Maps construction, application, and evaluation.

Project Experience

Predictive Display model and GPU code acceleration @ **MathWorks** (MATLAB, CUDA, Transformer)

- Proposed transformer model for XR teleoperation and achieved 6.6X speed up on Jetson Nano.

Real-Time Linux Kernel @ **RTEN Lab, UCR** (C, Scheduler, Real-time Clock, Linux Kernel)

- Developed IOCTL, LKM, High-resolution Timer, and TCB for real-time support and maintained high QoS and Timeliness Correctness of 1k+ concurrent tasks.

Academic Service

Reviewer @ Transactions on Embedded Computing Systems (TECS) 2025

Reviewer @ Journal of Systems Architecture (JSA) 2025

Secondary Reviewer @ Association for Computational Linguistics (ACL) 2025

External Reviewer @ IEEE Real-Time Systems Symposium (RTSS) 2024, 2025

External Reviewer @ ACM International Conference on Embedded Software (EMSOFT) 2024

External Reviewer @ Euromicro Conference on Real-Time Systems (ECRTS) 2024

Teaching Assistant @ Embedded Systems (EE128) 2023, 2024, 2025

Teaching Assistant @ Real-Time Embedded Systems (EE255) 2024

Teaching Assistant @ Computing Thinking (EDS124BR) 2019

Honors and Awards

Outstanding Teaching Award, UCR 2025

Conference Travel Grant, UCR 2024

Dean's Distinguished Fellowship Award, UCR 2022

Provost Honor, UCSD 2016, 2018

Technical Skills

Machine Learning: Tensorflow, PyTorch, ONNX, RecSys, RLHF, Distillation, RAG, Retrieval, Search;

GPU Programming: CUDA, CUDNN, GPU codegen/compiler, Kernel Scheduler, CUDA Graph;

Distributed Systems: AWS (EC2, ECS, DynamoDB, Lambda), LLM serving, Collaborative Inference;

On-device ML: NAS, FreeRTOS, LKM, Real-time Scheduling, ROS2, Networking, efficient LLM;

Programming Language: C/C++, Python, Java 8/11, Shell, HTML5/CSS3/Javascript, Typescript;

CI/CD: Perforce, Git, JIRA, Scrum, Kibana, Kubernetes, Docker, Postman;